

Supplementary Materials for “Estimating Heterogeneous Treatment Effects for General Responses” by Zijun Gao and Trevor Hastie

Overview. In Section 1, we discuss related works. In Section 2, we discuss variants and extensions of our proposed method. In Section 3, we provide proofs of the results in Section 3 and Section 4 of the main text. In Section 4, we include additional algorithms. In Section 5, we include details of the empirical study in the main text, as well as additional empirical experiments based on simulated or real data.

1. Literature

There is a rich literature on using flexible modeling techniques to estimate heterogeneous treatment effects. (Tian et al., 2014; Imai et al., 2013) formulate the estimation of heterogeneous treatment effects as a variable selection problem and consider a LASSO-type approach. (Athey and Imbens, 2015; Su et al., 2009; Foster et al., 2011) design recursive partitioning methods for causal inference and (Hill, 2011) adapts the Bayesian Additive Regression Trees (BART) to the causal setting. Ensemble learners, such as random forests (Wager and Athey, 2018) and boosting (Powers et al., 2018), have been investigated under the counterfactual framework. In addition, neural network-based causal estimators have also been proposed (Künzel et al., 2018; Shalit et al., 2017).

More recently, meta-learners for heterogeneous treatment effect estimation are of increasing popularity, which decompose the estimation task into sub-problems that can be solved by off-the-shelf machine learning tools (base learners) (Künzel et al., 2019). One common meta-algorithm, which we call separate estimation (SE) later, applies base learners to the treatment and control groups separately and then takes the difference (Foster et al., 2011;

(Lu et al., 2018; Hu et al., 2021; Foster et al., 2011). Another approach regards the treatment assignment as a covariate and uses base learners to learn the dependence on the enriched set of covariates. (Künzel et al., 2019) propose X-learner that first estimates the control group mean function, subtracts the predicted control counterfactuals from the observed treated responses, and finally estimates treatment effects from the differences. X-learner is effective if the control group and the treatment group are unbalanced in sample size. Robinson’s method (Robinson, 1988) and R-learner (Nie and Wager, 2021), also discussed in (Zhao et al., 2022), estimate the propensity score and the marginal mean function using arbitrary machine learning algorithms, and then minimizes a designed loss with plugged-in nuisance function estimators to learn the treatment effect. Robinson’s method and R-learner enjoy the robustness property to nuisance function estimators and admits flexible generalization through using alternative losses (more detailed discussion in Section 2 of the main text). DR-learner, as analyzed in (Kennedy, 2023), constructs pseudo-outcomes for the CATE and regresses them on the covariates. However, pseudo-outcomes for DINA cannot be constructed in the same way due to the nonlinear transformation from the original scale to the natural parameter scale. Explicitly, the transformed conditional expectation (e.g., $\log(\mathbb{E}[Y \mid X])$) — our target, differs from the conditional expectation of the transformed outcomes used by the pseudo-outcomes (e.g., $\mathbb{E}[\log(Y) \mid X]$).

2. Variants and extensions of the proposal

2.1 Extension to multiple-level treatment

In this section, we extend our method to address multiple-level treatments. We first introduce a few notations. Let $T \in \{0, 1, \dots, K\}$ be the treatment variable of $K + 1$ levels, and let level 0 denote the control group. Let $W^{(t)}$ be the indicator of receiving treatment

t , i.e., $W^{(t)} = \mathbb{1}_{\{T=t\}}$. We adopt the generalized propensity score of (Imbens, 2000),

$$e_t(x) = \mathbb{P}(T = t \mid X = x) = \mathbb{E}[W^{(t)} \mid X = x], \quad 0 \leq t \leq K. \quad (1)$$

We denote the natural parameter under treatment t by $\eta_t(x)$, and we aim to estimate the DINAs

$$\tau_t(x) := \eta_t(x) - \eta_0(x), \quad 1 \leq t \leq K. \quad (2)$$

The analysis with multiple-level treatments is reminiscent of that with single treatments.

We aim to rewrite the natural parameters as

$$\boldsymbol{\eta}(x) = \nu(x)\mathbf{1}_K + (\mathbf{I}_K - \mathbf{1}_K\mathbf{a}(x)^\top)\boldsymbol{\tau}(x), \quad (3)$$

where $\boldsymbol{\eta}(x) = [\eta_1(x), \dots, \eta_K(x)]^\top$, $\mathbf{a}(x) = [a_1(x), \dots, a_K(x)]^\top$, $\boldsymbol{\tau}(x) = [\tau_1(x), \dots, \tau_K(x)]^\top$, and $\mathbf{1}_K = [1, \dots, 1]^\top$. We look for $\mathbf{a}(x)$ and $\nu(x)$ such that the DINA estimator is not heavily disturbed by inaccurate nuisance functions. Analogous to Eq. (12) of the main text, we arrive at

$$\begin{aligned} a_t(x) &= \frac{e_t(x)V_t(x)}{\sum_{s=0}^K e_s(x)V_s(x)}, \quad 1 \leq t \leq K, \\ a_0(x) &= 1 - \sum_{t=1}^K a_t(x), \\ \nu(x) &= \sum_{t=0}^K a_t(x)\eta_t(x). \end{aligned}$$

The weight $a_t(x)$ is proportional to the treatment probability $e_t(x)$ multiplied by the response conditional variance. The multiplier $(w^{(t)} - a_t(x))$ associated with the HTE is small under treatment level t if the unit is inclined to receive such treatment or its treated response has high variance at x . Consequently, the proposed method equalizes the influences of treatments from all levels and attenuates the effects of excessively influential samples.

The algorithms in Section 3 of the main text can be directly generalized to handle multiple-level treatments. As for nuisance functions, we estimate the generalized propensity score $e_t(x)$ by any multi-class classifier that provides probability estimates. We estimate $V_t(x)$ by obtaining $\hat{\eta}_t(x)$ first and plugging it in to the appropriate exponential family formula for the variance.

Theoretically, Proposition 1 of the main text still holds, meaning that the proposed estimator will improve over separate estimation based on the nuisance-function estimators $\hat{\eta}_t(x)$.

Compared to single-level treatments, generalized propensity scores of multiple-level treatments may take more extreme values, especially values close to zero. Extreme propensities will cause inverse-propensity-weighting-type methods (Dudík et al., 2011; Weisberg and Pontes, 2015) to be highly variant; in contrast, while the variances of R-learner as well as our extension will not explode. We remark that when $\hat{\tau}(x)$ is estimated with a large number of parameters, problematic extrapolations may happen in the regions absent from certain treatment groups.

2.2 Cox model with full likelihood

In the main text, we have defined the baseline $\nu(x)$ and the modified propensity score $a(x)$ as

$$a(x) = \frac{e(x)\mathbb{P}(C \geq Y \mid W = 1, X = x)}{e(x)\mathbb{P}(C \geq Y \mid W = 1, X = x) + \{1 - e(x)\}\mathbb{P}(C \geq Y \mid W = 0, X = x)}, \quad (4)$$

$$\nu(x) = a(x)\eta_1(x) + \{1 - a(x)\}\eta_0(x).$$

We compare the nuisance-function construction (4) with that of the direct extension of R-learner and the DINA estimator for the exponential family. Different from R-learner, the $a(x)$ in (4) also relies on the censoring probabilities under treatment and control. For a unit with covariate value x , the multiplier $w - a(x)$ associated with the HTE is smaller for

$w = 1$ compared to $w = 0$ if the unit tends to be censored under treatment. Consequently, the weight $a(x)$ emphasizes more the units not censored, which agrees with the common sense that censored units contain less information. Compared with the nuisance function configuration in Section 3 of the main text, (4) does not include the derivatives $\frac{d\mu}{d\eta}(\eta_w(x))$ due to different parameter scales: in the Cox model, we focus on the tilting functions $\eta_w(x)$ instead of the natural parameters $e^{\eta_w(x)}$ in the exponential family.

We discuss several special instances of (4). In particular, we consider the subcases that the censoring times satisfy $C(0) \mid X \stackrel{d}{=} C(1) \mid X$, i.e., the conditional distributions of the censoring times are the same across the treatment and control arms.

- (1) No treatment effect. If there is no treatment effect, then the distributions of Y and C are conditionally independent of the treatment assignment indicator W . As a result, the probability ratio of not being censored is one and $a(x) = e(x)$ — R-learner.
- (2) No or light censoring. If the proportion of censored units is zero or small, the censoring time will be longer than the survival time despite the treatment effect, suggesting the ratio of not being censored to be close to one. Consequently, the case approximately reduces to R-learner.
- (3) Heavy censoring. If the proportion of censored units is high, then as argued by (Lin et al., 2013), almost all the information is contained in the indicator Δ , and we can directly model the rare event — “not censored” by Poisson distribution. In this case,

$$\frac{\mathbb{P}(C \geq Y \mid W = 1, X = x)}{\mathbb{P}(C \geq Y \mid W = 0, X = x)} \xrightarrow{\text{c.p.} \rightarrow 1} e^{\tau(x)}, \quad (5)$$

and plug (5) in (4) gives the $a(x)$ and $\nu(x)$ corresponding to the configuration with Poisson responses.

⁰c.p. stands for censored probability.

The following proposition demonstrates that the full-likelihood-based approach is robust for arbitrary $\tau(x)$.

PROPOSITION 1: Under the conditions of Proposition 1, Assumption 1, Assumption 2 of the main text, and assume the baseline hazard function $\lambda(y)$ is known, $\|a_n(x) - a(x)\|_2$, $\|\nu_n(x) - \nu(x)\|_2 = O(c_n)$, $c_n \rightarrow 0$, then the proposed estimator based on full likelihood satisfies

$$\|\beta_n - \beta\|_2 = \tilde{O}(c_n^2 + n^{-1/2}).$$

Proposition 1 is a stronger result compared to that of the partial-likelihood-based approach (Proposition 2 of the main text). Explicitly, the partial likelihood approach in Proposition 2 requires a null treatment effect assumption but does not require the knowledge of the true baseline hazard function.

The implementation details are summarized in Algorithm 1. Algorithm 1 is reminiscent of Algorithm 1 of the main text, except for the estimation of $a(x)$, which, in the survival setting, additionally requires estimating the censoring probabilities. We list several common random censoring mechanisms and the associated probabilities of “not censored” in Table 1. If the censoring mechanism is known, for example, all the units are enrolled simultaneously and operate for the same amount of time (singly-censored), then we can compute the censoring probabilities in closed form or via numerical integration given rough nuisance-function estimators $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$. If the censoring mechanism is unknown, we can directly estimate the censoring probabilities from the data: solving the classification problem with the response Δ , predictors (X, W) by any classifier that comes with predicted class probabilities. If the censoring is extremely low or high, we can turn to the approximations elaborated in the aforementioned special instances.

[Table 1 about here.]

3. Proofs

In this section, we present the proofs of the theoretical results in the main text. Section 3.1 contains the proof of Proposition 1 from Section 3 on exponential family. Section 3.2 provides the proofs of Propositions 2 and 3 from Section 4 on survival outcomes.

3.1 Proof of results for exponential family

3.1.1 Proof of Proposition 1 of the main text.

Proof. [Proof of Proposition 1 of the main text]

We first show the estimator is consistent, then prove the desired convergence rate, and finally turn to the efficiency statement.

Rewrite $a_n(x) - a(x) = \alpha_n \xi_n(x)$, $\nu_n(x) - \nu(x) = \rho_n \zeta_n(x)$, where $\mathbb{E}[\xi_n^2(X)]$ and $\mathbb{E}[\zeta_n^2(X)]$ are one. By the assumption that $a_n(x)$, $\nu_n(x)$ converge at rate $O(c_n)$, we have $\alpha_n, \rho_n = O(c_n)$. Let $s(a(X_i), \nu(X_i), \beta)$ denote the score function of the i -th unit, and define the empirical score $s_n(a(x), \nu(x), \beta) = \frac{1}{n} \sum_{i=1}^n s(a(X_i), \nu(X_i), \beta)$. For simplicity, we write $s_n(a_n(x), \nu_n(x), \beta_n)$ as $s_n(\alpha_n, \rho_n, \beta_n)$, then the derivatives of the score take the form

$$\begin{aligned}\nabla_{\alpha} s_n(\alpha_n, \rho_n, \beta_n) &= \frac{1}{n} \sum_{i=1}^n \nabla_{\alpha} s(a(X_i) + \alpha \xi_n(X_i), \nu(X_i) + \rho \zeta_n(X_i), \beta_n)|_{\alpha=\alpha_n}, \\ \nabla_{\rho} s_n(\alpha_n, \rho_n, \beta_n) &= \frac{1}{n} \sum_{i=1}^n \nabla_{\rho} s(a(X_i) + \alpha \xi_n(X_i), \nu(X_i) + \rho \zeta_n(X_i), \beta_n)|_{\rho=\rho_n}, \\ \nabla_{\beta} s_n(\alpha_n, \rho_n, \beta_n) &= \frac{1}{n} \sum_{i=1}^n \nabla_{\beta} s(a(X_i) + \alpha \xi_n(X_i), \nu(X_i) + \rho \zeta_n(X_i), \beta)|_{\beta=\beta_n}.\end{aligned}$$

Similarly we have the second order derivatives.

Consistency of β_n . Taylor's expansion of $s_n(\alpha_n, \rho_n, \beta_n)$ at $\alpha_n = \rho_n = 0$ is

$$s_n(\alpha_n, \rho_n, \beta_n) = s_n(0, 0, \beta_n) + \nabla_{\alpha} s_n(\alpha_{\varepsilon}, \rho_{\varepsilon}, \beta_n) \alpha_n + \nabla_{\rho} s_n(\alpha_{\varepsilon}, \rho_{\varepsilon}, \beta_n) \rho_n, \quad (6)$$

where $\alpha_\varepsilon \in [0, \alpha_n]$, $\rho_\varepsilon \in [0, \rho_n]$. Furthermore, Taylor's expansion of $s_n(0, 0, \beta_n)$ at the true β is

$$s_n(0, 0, \beta_n) = s_n(0, 0, \beta) + \nabla_\beta s_n(0, 0, \beta_\varepsilon)(\beta_n - \beta), \quad (7)$$

where $\beta_\varepsilon \in [\beta_n, \beta]$. By central limit theorem (CLT),

$$s_n(0, 0, \beta) = s(0, 0, \beta) + O_p(n^{-1/2}) = O_p(n^{-1/2}). \quad (8)$$

Under the regularity condition, $s_n(0, 0, \beta)$ is Lipschitz in β and $\mathcal{B}$ is bounded, then by the uniform law of large numbers (ULLN),

$$\sup_{\beta' \in \mathcal{B}} |\nabla_\beta s_n(0, 0, \beta') - \nabla_\beta s(0, 0, \beta')| = o_p(1). \quad (9)$$

Thus for n large enough, the minimal eigenvalue of $\nabla_\beta s_n(0, 0, \beta_\varepsilon)$ is at least half of the minimal eigenvalue of $\nabla_\beta s(0, 0, \beta_\varepsilon)$ and is lower bounded by $C/2$. Notice that $\nabla_\alpha s_n(\alpha_n, \rho_n, \beta_n)$, $\nabla_\rho s_n(\alpha_n, \rho_n, \beta_n)$ are bounded, we plug Eq. (7), (8), (9) in Eq. (6),

$$\begin{aligned} 0 &= s_n(\alpha_n, \rho_n, \beta_n) = \nabla_\beta s_n(0, 0, \beta_\varepsilon)(\beta_n - \beta) + O_p(n^{-1/2} + \alpha_n + \rho_n) \\ \implies \beta_n - \beta &= (\nabla_\beta s_n(0, 0, \beta_\varepsilon))^{-1} O_p(n^{-1/2} + \alpha_n + \rho_n) = o_p(1). \end{aligned}$$

Therefore, β_n is consistent.

Convergence rate of β_n . The second order Taylor's expansion of $s_n(\alpha_n, \rho_n, \beta_n)$ at $\beta_n = \beta$ and $\alpha = \rho = 0$ is

$$\begin{aligned} s_n(\alpha_n, \rho_n, \beta_n) &= s_n(0, 0, \beta_n) + \nabla_\alpha s_n(0, 0, \beta_n)\alpha_n + \nabla_\rho s_n(0, 0, \beta_n)\rho_n \\ &\quad + \frac{1}{2}\nabla_\alpha^2 s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\alpha_n^2 + \frac{1}{2}\nabla_\rho^2 s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\rho_n^2 + \nabla_{\alpha\rho} s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\alpha_n\rho_n \\ &= s_n(0, 0, \beta) + \nabla_\beta s_n(0, 0, \beta_\varepsilon)(\beta_n - \beta) + \nabla_\alpha s_n(0, 0, \beta_n)\alpha_n + \nabla_\rho s_n(0, 0, \beta_n)\rho_n \\ &\quad + \frac{1}{2}\nabla_\alpha^2 s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\alpha_n^2 + \frac{1}{2}\nabla_\rho^2 s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\rho_n^2 + \nabla_{\alpha\rho} s_n(\alpha_\varepsilon, \rho_\varepsilon, \beta_n)\alpha_n\rho_n, \end{aligned} \quad (10)$$

where $\beta_\varepsilon \in [\beta_n, \beta]$, $\alpha_\varepsilon \in [0, \alpha]$, $\rho_\varepsilon \in [0, \rho]$. The first order Taylor's expansion of $\nabla_\alpha s_n(0, 0, \beta_n)$

at β is

$$\begin{aligned}
\nabla_{\alpha} s_n(0, 0, \beta_n) &= \nabla_{\alpha} s_n(0, 0, \beta) + \nabla_{\alpha\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta) \\
&= \nabla_{\alpha} s(0, 0, \beta) + O_p(n^{-1/2}) + \nabla_{\alpha\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta) \\
&= O_p(n^{-1/2}) + \nabla_{\alpha\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta),
\end{aligned} \tag{11}$$

where we apply CLT again in the second last equation, and use $\nabla_{\alpha} s(0, 0, \beta) = 0$ in the last equation. Similar analysis can be done for $\nabla_{\rho} s_n(0, 0, \beta_n)$. Plug Eq. (8), (11) into Eq. (10),

$$\begin{aligned}
s_n(\alpha_n, \rho_n, \beta) &= O_p(n^{-1/2}) + \nabla_{\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta) + O_p(n^{-1/2}(\alpha_n + \rho_n)) \\
&\quad + \nabla_{\alpha\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta)\alpha_n + \nabla_{\rho\beta} s_n(0, 0, \beta_{\varepsilon})(\beta_n - \beta)\rho_n \\
&\quad + \frac{1}{2}\nabla_{\alpha}^2 s_n(\alpha_{\varepsilon}, \rho_{\varepsilon}, \beta_n)\alpha_n^2 + \frac{1}{2}\nabla_{\rho}^2 s_n(\alpha_{\varepsilon}, \rho_{\varepsilon}, \beta_n)\rho_n^2 + \nabla_{\alpha\rho} s_n(\alpha_{\varepsilon}, \rho_{\varepsilon}, \beta_n)\alpha_n\rho_n \\
&= \{\nabla_{\beta} s_n(0, 0, \beta_{\varepsilon}) + O_p(\alpha_n + \rho_n)\}(\beta_n - \beta) + O_p(\alpha_n^2 + \rho_n^2 + \alpha_n\rho_n + n^{-1/2}),
\end{aligned} \tag{12}$$

where we use the condition that the second derivatives are bounded. As shown above, the minimal eigenvalue of $\nabla_{\beta} s_n(0, 0, \beta_{\varepsilon})$ is uniformly lower bounded by $C/2$. Thus Eq. (12) implies

$$\beta_n - \beta = \{\nabla_{\beta} s_n(0, 0, \beta_{\varepsilon}) + O_p(\alpha_n + \rho_n)\}^{-1} O_p(n^{-1/2} + \alpha_n^2 + \rho_n^2 + \alpha_n\rho_n) = O_p(n^{-1/2} + c_n^2),$$

and we finish the proof.

Efficiency. Let $\tilde{\eta}_0(x)$ be the control group baseline function, $\tilde{\tau}(x)$ be some treatment effect function, and $\ell_i(\tilde{\tau}(x); \tilde{\eta}_0(x))$ take the form of the log likelihood from the exponential family suitable for the response type, evaluated at the i -th observation (X_i, W_i, Y_i) ,

$$\ell_i(\tilde{\tau}(x); \tilde{\eta}_0(x)) = Y_i(\tilde{\eta}_0(X_i) + W_i\tilde{\tau}(X_i)) - \psi(\tilde{\eta}_0(X_i) + W_i\tilde{\tau}(X_i)). \tag{13}$$

Then the maximizer of the expectation of ℓ_i is, under unconfoundedness,

$$\begin{aligned}\eta_0(x) &= (\psi')^{-1}(\mathbb{E}[Y(0) \mid X = x]), \\ \tau(x) &= (\psi')^{-1}(\mathbb{E}[Y(1) \mid X = x]) - (\psi')^{-1}(\mathbb{E}[Y(0) \mid X = x]).\end{aligned}$$

This implies that $\hat{\tau}(x)$ converges to the difference in transformed conditional expectations between the treatment and control groups, even with misspecified model. This also indicates that taking the derivative of ℓ_i with respect to $\tau(x)$, or with respect to β when $\tau(x) = x^\top \beta$, yields a proper score function. Based on this, we derive the Neyman orthogonal score for $\tau(x)$ using the concentrating-out approach (Newey, 1994), and show that the score function associated with the likelihood with our parametrization $(\nu(x), a(x))$ recovers the Neyman orthogonal score. As a result, our proposal inherits the desirable properties of an estimator obtained from the concentrating-out approach regarding local doubly-robustness and the efficiency.

Derivation of the Neyman orthogonal score using the concentrating-out approach. The following derivation is valid regardless of whether the model of (X_i, W_i, Y_i) is well-specified. We consider estimating the treatment effect by a linear function of the covariates, i.e., $\tilde{\tau}(x) = x^\top \tilde{\beta}$, $\tilde{\beta} \in \mathcal{B}$, and $\mathcal{B}$ is a bounded convex subset of $\mathbb{R}^d$. The derivation can be readily extended to estimating the treatment effect by a linear combination of a series of pre-specified basis functions, like natural cubic splines.

Given any $\tilde{\tau}(x) = x^\top \tilde{\beta}$, we define the associated optimal nuisance function as

$$\eta_0(x; \tilde{\beta}) = \operatorname{argmax}_{\tilde{\eta}_0(x)} \mathbb{E} \left[\ell_i(x^\top \tilde{\beta}; \tilde{\eta}_0(x)) \right]. \quad (14)$$

For any x in the domain of X_i , consider $\eta_0(x; \tilde{\beta}) + t\delta_{\{X_i=x\}}$, where $\delta_{\{X_i=x\}}$ is the Dirac

function at x , then (14) implies

$$\begin{aligned}
0 &= \partial_t \mathbb{E} \left[\ell_i \left(x^\top \tilde{\beta}; \tilde{\eta}_0(x; \tilde{\beta}) + t \delta_{\{X_i=x\}} \right) \right] \Big|_{t=0} \\
&= \mathbb{E} \left[\partial_t \ell_i \left(x^\top \tilde{\beta}; \tilde{\eta}_0(x; \tilde{\beta}) + t \delta_{\{X_i=x\}} \right) \right] \Big|_{t=0} \\
&= \mathbb{E} \left[\left\{ Y_i - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta}) \right\} \delta_{\{X_i=x\}} \right] \\
&= \mathbb{E} \left[Y_i - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta}) \mid X_i = x \right] \\
&= \mathbb{E} \left[Y_i - \psi'(\eta_0(x; \tilde{\beta}) + W_i x^\top \tilde{\beta}) \mid X_i = x \right].
\end{aligned} \tag{15}$$

Here we assume the regularity to enable the exchange of the expectation and the partial derivative ∂_t . We further take derivative with regard to $\tilde{\beta}$ on both sides of (15),

$$\begin{aligned}
0 &= \partial_{\tilde{\beta}} \mathbb{E} [Y_i - \psi'(\eta_0(x; \tilde{\beta}) + W_i x^\top \tilde{\beta}) \mid X_i = x] \\
&= -\mathbb{E} [\psi''(\eta_0(x; \tilde{\beta}) + W_i x^\top \tilde{\beta}) \cdot \{\partial_{\tilde{\beta}} \eta_0(x; \tilde{\beta}) + W_i x\} \mid X_i = x] \\
&= -\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot \{\partial_{\tilde{\beta}} \eta_0(x; \tilde{\beta}) + x\} \cdot e(x) \\
&\quad - \psi''(\eta_0(x; \tilde{\beta})) \cdot \partial_{\tilde{\beta}} \eta_0(x; \tilde{\beta}) \cdot \{1 - e(x)\} \\
&= -\partial_{\tilde{\beta}} \eta_0(x; \tilde{\beta}) \left[\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) + \psi''(\eta_0(x; \tilde{\beta})) \cdot \{1 - e(x)\} \right] \\
&\quad - \psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) \cdot x.
\end{aligned} \tag{16}$$

Here we again assume sufficient regularity such that we can swap the order of the expectation and the derivative. Eq. (16) gives us

$$\partial_{\tilde{\beta}} \eta_0(x; \tilde{\beta}) = -\frac{\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) \cdot x}{\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) + \psi''(\eta_0(x; \tilde{\beta})) \cdot \{1 - e(x)\}} = -a(x) \cdot x, \tag{17}$$

which yields the $a(x)$ used in our proposal.

Next, we plug $\eta_0(x; \tilde{\beta})$ into the objective function (13), and maximize it to estimate β ,

$$\max_{\tilde{\beta} \in \mathcal{B}} \ell_i(x^\top \tilde{\beta}; \eta_0(x; \tilde{\beta})). \tag{18}$$

The objective function (18) is a function of $\tilde{\beta}$. We compute the associated score function,

$$\begin{aligned}
s_i(x^\top \tilde{\beta}; \eta_0(x; \tilde{\beta})) &= \partial_{\tilde{\beta}} \ell_i(x^\top \tilde{\beta}; \eta_0(x; \tilde{\beta})) \\
&= Y_i \cdot W_i X_i - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta}) \cdot W_i X_i \\
&\quad + Y_i \cdot \partial_{\tilde{\beta}} \eta_0(X_i; \tilde{\beta}) - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta}) \cdot \partial_{\tilde{\beta}} \eta_0(X_i; \tilde{\beta}) \\
&= \{Y_i - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta})\} \cdot \{W_i X_i + \partial_{\tilde{\beta}} \eta_0(X_i; \tilde{\beta})\}.
\end{aligned} \tag{19}$$

Plug the derivative $\partial_{\tilde{\beta}} \eta_0(X_i; \tilde{\beta})$ in Eq. (17) into the score (19) and we obtain the Neyman orthogonal score

$$s_i(x^\top \tilde{\beta}; \eta_0(x; \tilde{\beta})) = \{Y_i - \psi'(\eta_0(X_i; \tilde{\beta}) + W_i X_i^\top \tilde{\beta})\} \cdot \{W_i X_i - a(X_i) X_i\}. \tag{20}$$

Below we show that the score (20) is equivalent to the score function derived from our proposal with nuisance functions $a(x)$, $\nu(x)$. In our proposal, we investigate the baseline function $\nu(x)$ that is a mixture of the natural parameters

$$\nu(x) = a(x) \eta_1(x) + \{1 - a(x)\} \eta_0(x), \tag{21}$$

where the associated modified propensity score is defined as ¹

$$a(x) = \frac{\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) \cdot x}{\psi''(\eta_0(x; \tilde{\beta}) + x^\top \tilde{\beta}) \cdot e(x) + \psi''(\eta_0(x; \tilde{\beta})) \cdot \{1 - e(x)\}} \tag{22}$$

Provided with $\tilde{a}(x)$, $\tilde{\nu}(x)$, we rewrite the likelihood based on (21), (22) and obtain the expression

$$\ell_i(x^\top \tilde{\beta}; \tilde{a}(x), \tilde{\nu}(x)) := Y_i \{\tilde{\nu}(X_i) + (W_i - \tilde{a}(X_i)) X_i^\top \tilde{\beta}\} - \psi \left(\tilde{\nu}(X_i) + \{W_i - \tilde{a}(X_i)\} X_i^\top \tilde{\beta} \right). \tag{23}$$

¹In the manuscript, we introduce the abbreviation $V_1(x) = \frac{d\mu}{d\eta}(\eta_1(x; \tilde{\beta})) = \psi''(\eta_1(x; \tilde{\beta}))$ and $V_0(x) = \frac{d\mu}{d\eta}(\eta_0(x; \tilde{\beta})) = \psi''(\eta_0(x; \tilde{\beta}))$ and write $a(x) = \frac{e(x)V_1(x)}{e(x)V_1(x) + (1-e(x))V_0(x)}$. There we use V to denote the second derivative because, when the data is actually drawn from the exponential family, $\psi''(\eta_0(x; \tilde{\beta}) + w x^\top \tilde{\beta})$ is the conditional ‘‘V’’ariance of Y given $X = x$, $W = w$.

Take the derivative with regard to $\tilde{\beta}$ and we arrive at the score function

$$\left\{ Y_i - \psi' \left(\tilde{\nu}(X_i) + \{W_i - \tilde{a}(X_i)\} X_i^\top \tilde{\beta} \right) \right\} \{W_i - \tilde{a}(X_i)\} X_i. \quad (24)$$

Based on (21), (22), the score (24) is the same as the score (20) derived using the concentrating-out approach.

Note that the score derived from the concentrating-out approach is efficient if and only if the likelihood is correctly specified, which yields the desired result.

We also remark that the derivation does not require the responses to be drawn from an exponential family model.

3.2 Proof of results for Cox model

3.2.1 Proof of Proposition 1 of the Supplementary Materials.

Proof. [Proof of Proposition 1 of the Supplementary Materials] To show Proposition 2 of the main text is true, we only need to verify the score function is Neyman's orthogonal, and the rest of the proof is the same as that of Proposition 1 of the main text.

The score equation of the full likelihood function at $a_n(x) = a(x) + \alpha_n \xi_n(x)$, $\nu_n(x) = \nu(x) + \rho_n \zeta_n(x)$ is

$$s(\alpha_n, \rho_n, \beta_n) = \mathbb{E} \left[\Delta \{W - a_n(X)\} X - \Lambda(Y^c) \{W - a_n(X)\} X e^{\nu_n(X) + \{W - a_n(X)\} X^\top \beta_n} \right]. \quad (25)$$

The derivative of (25) with regard to ρ evaluated at $\alpha = \rho = 0$ and $\beta_n = \beta$ is

$$\nabla_\rho s(0, 0, \beta) = -\mathbb{E} \left[\Lambda(Y^c) \{W - a(X)\} X e^{\nu(X) + \{W - a(X)\} X^\top \beta} \zeta(X) \right]. \quad (26)$$

By Lemma 1, (26) becomes

$$\begin{aligned}\nabla_{\rho}s(0, 0, \beta) &= -\mathbb{E} \left[\mathbb{P}(C \geq Y \mid W, X) e^{-\eta_w(X)} \{W - a(X)\} X e^{\nu(X) + \{W - a(X)\} X^{\top} \beta} \zeta(X) \right] \\ &= -\mathbb{E} [\mathbb{P}(C \geq Y \mid W, X) \{W - a(X)\} X \zeta(X)] = 0.\end{aligned}$$

Next, the derivative of (25) with regard to α evaluated at $\alpha = \rho = 0$ and $\beta_n = \beta$ is

$$\begin{aligned}\nabla_{\alpha}s(0, 0, \beta) &= -\mathbb{E} [\Delta \xi(X) X] + \mathbb{E} \left[\Lambda(Y^c) \xi(X) X e^{\nu(X) + \{W - a(X)\} X^{\top} \beta} \right] \\ &\quad + \mathbb{E} \left[\Lambda(Y^c) \{W - a(X)\} X e^{\nu(X) + \{W - a(X)\} X^{\top} \beta} \xi(X) X^{\top} \beta \right].\end{aligned}\tag{27}$$

The third term in (27) is zero by the same argument of (26). For the second term, by Lemma 1,

$$\mathbb{E} \left[\Lambda(Y^c) \xi(X) X e^{\nu(X) + \{W - a(X)\} X^{\top} \beta} \right] = \mathbb{E} [\mathbb{P}(C \geq Y \mid W, X) \xi(X) X] = \mathbb{E} [\Delta \xi(X) X].$$

Therefore, combine the three terms and we prove (27) is zero.

In this way, we have verified the Neyman's orthogonality of the full likelihood score with known baseline hazard.

3.2.2 Proof of Proposition 2 of the main text.

Proof. [Proof of Proposition 2 of the main text]

To show Proposition 2 of the main text is true, we only need to verify the asymptotic score is Neyman's orthogonal if $\tau(x) = 0$, and the rest of the proof is the same as Proposition 1.

According to (Tsiatis, 1981), the score equation of the partial likelihood function at $a_n(x) = a(x) + \alpha_n \xi_n(x)$, $\nu_n(x) = \nu(x) + \rho_n \zeta_n(x)$, and β_n is asymptotically

$$\begin{aligned}s(\alpha_n, \rho_n, \beta_n) &= \mathbb{E} [\Delta(W - a_n(X)) X] \\ &\quad - \int_0^{\infty} \frac{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu_n(X) + \{W - a_n(X)\} X^{\top} \beta_n} \{W - a_n(X)\} X \right]}{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu_n(X) + \{W - a_n(X)\} X^{\top} \beta_n} \right]} d\mathbb{P}(y),\end{aligned}\tag{28}$$

where $d\mathbb{P}(y) = \lambda(y) \mathbb{E} [e^{\eta_w(X)} 1_{\{Y^c \geq y\}}] dy$ with the true parameter $\eta_w(x)$. The derivative of

the score (28) with regard to ρ evaluated at $\alpha = \rho = 0$ and $\beta_n = \beta$ is

$$\begin{aligned} \nabla_\rho s(0, 0, \beta) &= \underbrace{\int_0^\infty \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \zeta(X) \{W - a(X)\} X \right] \lambda(y) dy}_{:= (I)} \\ &\quad - \underbrace{\int_0^\infty \frac{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \{W - a(X)\} X \right] \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \zeta(X) \right]}{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \right]} \lambda(y) dy}_{:= (II)}. \end{aligned} \quad (29)$$

Part (I) is zero for arbitrary β by the analysis of (II.a) in (32).

As for part (II), recall the definition of $a(x)$ and that $\beta = 0$,

$$\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \{W - a(X)\} X \right] = \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X)} \{W - a(X)\} X \right] = 0.$$

Next, the derivative of the score (28) with regard to α evaluated at $\alpha = \rho = 0$, $\beta_n = \beta$ is

$$\begin{aligned} \nabla_\alpha s(\alpha, \rho, \beta) &= -\mathbb{E}[\Delta \xi(X) X] \\ &\quad + \underbrace{\int_0^\infty \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \xi(X) X^\top \beta \{W - a(X)\} X \right] \lambda(y) dy}_{:= (I'.a)} \\ &\quad + \underbrace{\int_0^\infty \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \xi(X) X \right] \lambda(y) dy}_{:= (I'.b)} \\ &\quad - \underbrace{\int_0^\infty \frac{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \{W - a(X)\} X \right] \mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \xi(X) X^\top \beta \right]}{\mathbb{E} \left[1_{\{Y^c \geq y\}} e^{\nu(X) + \{W - a(X)\} X^\top \beta} \right]} \lambda(y) dy}_{:= (II')}. \end{aligned} \quad (30)$$

The analysis of part (I'a) and (II') are the same as that of part (I) and (II) in (29). For part (I'b), for arbitrary β , the analysis of (32) shows

$$(I'b) = \mathbb{E}[\mathbb{P}(C \geq Y \mid W, X) \xi(X) X],$$

which cancels the term $-\mathbb{E}[\Delta \xi(X) X]$ in (30). Then combine all parts together and we have shown (30) is zero.

3.3 Proof of supporting lemmas

LEMMA 1: *Let Y be generated from the Cox model with the parameter function $\eta_w(x)$. Consider random censorship and let $f_C(c \mid W, X)$ be the conditional density of the censoring*

time. Then

$$\begin{aligned}\mathbb{E}[\Delta \mid W, X] &= \mathbb{P}(C \geq Y \mid W, X) = \int \left\{1 - e^{-\Lambda(c)e^{\eta_W(X)}}\right\} f_C(c \mid W, X) \, dc, \\ \mathbb{E}[\Lambda(Y^c) \mid W, X] &= e^{-\eta_W(X)} \int \left\{1 - e^{-\Lambda(c)e^{\eta_W(X)}}\right\} f_C(c \mid W, X) \, dc.\end{aligned}$$

Proof. [Proof of Lemma 1] By definition, the first equality in Lemma 1 is true. For the second equality, by Assumption 2 of the main text,

$$\begin{aligned}\mathbb{E}[\Lambda(Y^c) \mid W, X] &= \mathbb{E}[\Lambda(Y)\mathbb{1}_{\{C \geq Y\}} \mid W, X] + \mathbb{E}[\Lambda(C)\mathbb{1}_{\{C < Y\}} \mid W, X] \\ &= \int_0^\infty \int_0^c \Lambda(y)e^{-\Lambda(y)e^{\eta_W(X)}} d\Lambda(y)e^{\eta_W(X)} f_C(c \mid W, X) dc \\ &\quad + \int_0^\infty \int_c^\infty \Lambda(c)e^{-\Lambda(y)e^{\eta_W(X)}} d\Lambda(y)e^{\eta_W(X)} f_C(c \mid W, X) dc \\ &= e^{-\eta_W(X)} \int_0^\infty \int_0^{\Lambda(c)e^{\eta_W(X)}} ze^{-z} dz f_C(c \mid W, X) dc \\ &\quad + \int_0^\infty \Lambda(c)e^{-\Lambda(c)e^{\eta_W(X)}} f_C(c \mid W, X) dc \\ &= e^{-\eta_W(X)} \int_0^\infty \left[1 - e^{-\Lambda(c)e^{\eta_W(X)}} \{\Lambda(c)e^{\eta_W(X)} + 1\}\right] f_C(c \mid W, X) dc \\ &\quad + \int_0^\infty \Lambda(c)e^{-\Lambda(c)e^{\eta_W(X)}} f_C(c \mid W, X) dc \\ &= e^{-\eta_W(X)} \int_0^\infty \left\{1 - e^{-\Lambda(c)e^{\eta_W(X)}}\right\} f_C(c \mid W, X) dc,\end{aligned}$$

where we use $\int ze^{-z} dz = -e^{-z}(z + 1)$.

LEMMA 2: Let Y be generated from the Cox model with the natural parameter function $\eta_w(x) = \nu(x) + \{w - a(x)\}x^\top \beta$. Then for $\eta_w^*(x) = \nu^*(x) + \{w - a(x)\}x^\top \beta$ and $\eta'_w(x) =$

$\nu^*(x) + \{w - a(x)\}x^\top \beta'$, the asymptotic partial likelihood satisfies

$$\begin{aligned}
 pl(\eta'_w(x)) &= pl(\eta_w^*(x)) + (\beta' - \beta)^\top \left\{ \underbrace{\mathbb{E}[(\nu(X) - \nu^*(X))\mathbb{P}(C \geq Y \mid W, X)\{W - a(X)\}X]}_{:= (a)} \right. \\
 &\quad \left. \underbrace{\int_0^\infty \frac{\mathbb{E}[(\nu(X) - \nu^*(X))e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]}{\mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]} \mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}\{W - a(X)\}X] d\mathbb{P}(y)}_{:= (b)} \right\} \\
 &\quad - (\beta' - \beta)^\top \Sigma (\beta' - \beta) + O(\|\beta' - \beta\|_2^3),
 \end{aligned}$$

where Σ is a positively definite weight matrix.

Proof. [Proof of Lemma 2] Under Cox model, the asymptotic partial likelihood obeys

$$pl(\eta'_w(x)) = pl(\eta_w^*(x)) + \underbrace{\mathbb{E}[\Delta\{W - a(X)\}X^\top(\beta' - \beta)]}_{:= (I)} - \underbrace{\int_0^\infty \log \left(\frac{\mathbb{E}[e^{\eta'_w(X)}1_{\{Y^c \geq y\}}]}{\mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]} \right) d\mathbb{P}(y)}_{:= (II)}.$$

For part (I), by the random censoring mechanism,

$$(I) = \mathbb{E}[\mathbb{P}(C \geq Y \mid W, X)\{W - a(X)\}X^\top(\beta' - \beta)]. \quad (31)$$

For part (II), we apply the approximation $\log(1 + x) = x - x^2/2 + O(x^3)$ and get

$$\begin{aligned}
 (II) &= \underbrace{\int_0^\infty \frac{\mathbb{E}[(e^{\eta'_w(X)} - e^{\eta_w^*(X)})1_{\{Y^c \geq y\}}]}{\mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]} d\mathbb{P}(y)}_{:= (II.a)} \\
 &\quad - \underbrace{\int_0^\infty \frac{1}{2} \left(\frac{\mathbb{E}[(e^{\eta'_w(X)} - e^{\eta_w^*(X)})1_{\{Y^c \geq y\}}]}{\mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]} \right)^2 d\mathbb{P}(y)}_{:= (II.b)} + r,
 \end{aligned}$$

where the residual $r = O\left(\int_0^\infty \left(\frac{\mathbb{E}[(e^{\eta'_w(X)} - e^{\eta_w^*(X)})1_{\{Y^c \geq y\}}]}{\mathbb{E}[e^{\eta_w^*(X)}1_{\{Y^c \geq y\}}]}\right)^3 d\mathbb{P}(y)\right) = O(\|\eta'_w(X) - \eta_w^*(X)\|_2^3)$,

and thus is $O(\|\beta - \beta'\|_2^3)$. Notice that

$$\begin{aligned}
& \mathbb{E} \left[\left\{ e^{\eta'_W(X)} - e^{\eta_W^*(X)} \right\} 1_{\{Y^c \geq y\}} \right] \\
&= \mathbb{E} \left[\{ \eta'_W(X) - \eta_W^*(X) \} e^{\eta_W^*(X)} 1_{\{Y^c \geq y\}} \right] + O \left(\mathbb{E} \left[\{ \eta'_W(X) - \eta_W^*(X) \}^2 \right] \right) \\
&= \mathbb{E} \left[\{ W - a(X) \} X^\top (\beta' - \beta) e^{\eta_W^*(X)} 1_{\{Y^c \geq y\}} \right] + O \left(\mathbb{E} \left[\{ \eta'_W(X) - \eta_W^*(X) \}^2 \right] \right) \\
&= (\beta' - \beta)^\top \mathbb{E} \left[\{ W - a(X) \} X e^{\eta_W^*(X)} 1_{\{Y^c \geq y\}} \right] + O \left(\mathbb{E} \left[\{ \eta'_W(X) - \eta_W^*(X) \}^2 \right] \right),
\end{aligned}$$

thus (II.b) is approximately a quadratic term of $(\beta - \beta')$.

For part (II.a), notice that $d\mathbb{P}(y) = \lambda(y) \mathbb{E} [e^{\eta_W(X)} 1_{\{Y^c \geq y\}}]$,

$$\begin{aligned}
(\text{II.a}) &= \underbrace{\int_0^\infty \mathbb{E} \left[\left\{ e^{\eta'_W(X)} - e^{\eta_W^*(X)} \right\} 1_{\{Y^c \geq y\}} \right] \lambda(y) dy}_{:= (\text{II.a.1})} \\
&\quad + \underbrace{\int_0^\infty \frac{\mathbb{E} \left[\left\{ e^{\eta'_W(X)} - e^{\eta_W^*(X)} \right\} 1_{\{Y^c \geq y\}} \right] \mathbb{E} \left[\left\{ e^{\eta_W(X)} - e^{\eta_W^*(X)} \right\} 1_{\{Y^c \geq y\}} \right]}{\mathbb{E} [e^{\eta_W^*(X)} 1_{\{Y^c \geq y\}}]} d\mathbb{P}(y)}_{:= (\text{II.a.2})}.
\end{aligned}$$

We first compute the term (II.a.1). By the assumption of censoring and $\mathbb{P}(Y \geq y \mid X = x, W = w) = e^{-\Lambda(y)e^{-\eta_W(x)}}$, we exchange the order of expectation and integral,

$$\begin{aligned}
(\text{II.a.1}) &= \mathbb{E} \left[\int_0^\infty \left\{ e^{\eta'_W(X)} - e^{\eta_W^*(X)} \right\} 1_{\{Y \geq y\}} 1_{\{C \geq y\}} \lambda(y) dy \right] \\
&= \mathbb{E} \left[\left\{ e^{\eta'_W(X)} - e^{\eta_W^*(X)} \right\} \int_0^\infty \int_0^c e^{-\Lambda(y)e^{\eta_W(X)}} \lambda(y) dy f_C(c \mid W, X) dc \right] \\
&= \mathbb{E} \left[\left\{ e^{\eta'_W(X) - \eta_W^*(X)} - 1 \right\} e^{\eta_W^*(X) - \eta_W(X)} \int_0^\infty \int_0^c e^{-\Lambda(y)e^{\eta_W(X)}} d\Lambda(y) e^{\eta_W(X)} f_C(c \mid W, X) dc \right] \\
&= \mathbb{E} \left[\left\{ e^{\eta'_W(X) - \eta_W^*(X)} - 1 \right\} e^{\eta_W^*(X) - \eta_W(X)} \int_0^\infty \left\{ 1 - e^{-\Lambda(c)e^{\eta_W(X)}} \right\} f_C(c \mid W, X) dc \right] \\
&= \mathbb{E} \left[\left\{ e^{\eta'_W(X) - \eta_W^*(X)} - 1 \right\} e^{\eta_W^*(X) - \eta_W(X)} \mathbb{P}(C \geq Y \mid W, X) \right],
\end{aligned} \tag{32}$$

where we use Lemma 1 in the last equality. Since $\eta'_W(X) - \eta_W^*(X) = \{W - a(X)\}X^\top(\beta' - \beta)$,

$$\begin{aligned}
(\text{II.a.1}) &= \mathbb{E} \left[\{\eta'_W(X) - \eta_W^*(X)\} e^{\eta_W^*(X) - \eta_W(X)} \mathbb{P}(C \geq Y \mid W, X) \right] \\
&\quad + (\beta' - \beta)^\top \Sigma (\beta' - \beta) + O(\|\beta' - \beta\|_2^3) \\
&= \mathbb{E} \left[\{W - a(X)\} X^\top (\beta' - \beta) e^{\eta_W^*(X) - \eta_W(X)} \mathbb{P}(C \geq Y \mid W, X) \right] \\
&\quad + (\beta' - \beta)^\top \Sigma (\beta' - \beta) + O(\|\beta' - \beta\|_2^3).
\end{aligned} \tag{33}$$

Combine (II.a.1) in (33) and (I) in (31),

$$\begin{aligned}
(\text{I}) - (\text{II.a.1}) &= \mathbb{E} \left[\mathbb{P}(C \geq Y \mid W, X) \{W - a(X)\} X^\top (\beta' - \beta) \{1 - e^{\eta_W^*(X) - \eta_W(X)}\} \right] \\
&\quad - (\beta' - \beta)^\top \Sigma (\beta' - \beta) + O(\|\beta' - \beta\|_2^3) \\
&\approx \mathbb{E} \left[\mathbb{P}(C \geq Y \mid W, X) \{W - a(X)\} X^\top (\beta' - \beta) \{\nu(X) - \nu^*(X)\} \right] \\
&\quad - (\beta' - \beta)^\top \Sigma (\beta' - \beta) + O(\|\beta' - \beta\|_2^3).
\end{aligned} \tag{34}$$

In this way, (I) - (II.a.1) gives part (a) in Lemma 2. Furthermore, (II.a.2) gives part (b) in Lemma 2. Combine all parts and we finish the proof.

4. Additional algorithms

We present the algorithms for the Cox model with full likelihood (assuming a known baseline hazard function) in Algorithm 1, and with partial likelihood (unknown baseline hazard function) in Algorithm 2.

Algorithm 1: Cox model with full likelihood

Input: baseline hazard function $\lambda(y)$ (or the baseline cumulative hazard function $\Lambda(y)$).

We split the data randomly into two equal folds.

1. On fold one,

- (1) Estimation of $e(x)$.
- (2) Estimation of $\eta_0(x)$, $\eta_1(x)$. We maximize the full likelihood on the control and treatment group respectively provided with the hazard function $\lambda(y)$, and obtain estimators $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$.
- (3) Estimation of the probability of not being censored. We compute/estimate the not-censored probabilities $\hat{\mathbb{P}}(C \geq Y \mid X = x, W = w)$ based on closed form solution, by numerical integral, or by classifiers with response Δ and predictors (X, W) .
- (4) Substitution. We construct $\hat{a}(x)$ and $\hat{\nu}(x)$ based on $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$, $\hat{e}(x)$, and $\hat{\mathbb{P}}(C \geq Y \mid X = x, W = w)$.

2. On fold two, we plug in $\hat{a}(x)$ and $\hat{\nu}(x)$ to estimate $\tau(x)$ by maximizing the full likelihood

$$\max_{\tau'(x)} \ell(\tau'(x); \hat{a}(x), \hat{\nu}(x)) := \frac{1}{n} \sum_{i=1}^n \Delta_i [\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\} \tau'(X_i)] - \Lambda(Y_i^c) e^{\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\} \tau'(X_i)}. \quad (35)$$

3. We swap the folds and obtain another estimate. We average the two estimates and output.

5. Additional Empirical results

5.1 Simulation

5.1.1 *Description of simulation settings of the main text.* The specific configurations of the hyperparameters α , β , θ , and δ are provided in the supplementary materials as a separate Excel file *simulation setting.xlsx*.

5.1.2 *Description of real data analysis of the main text.* The SPRINT data (Powers et al., 2018) is collected from a randomized trial aiming to study whether a new treatment program targeting reducing systolic blood pressure (SBP) will reduce cardiovascular disease (CVD) risk. The response is whether any of the major CVD events² occur to a participant.

²Major CVD events: myocardial infarction (MI), non-MI acute coronary syndrome (non-MI ACS), stroke, heart failure (HF), or death attributable to cardiovascular disease.

Algorithm 2: Cox model with partial likelihood

We split the data randomly into two equal folds.

1. On fold one,

- (1) Estimation of $e(x)$.
- (2) Estimation of $\eta_0(x)$, $\eta_1(x)$. We maximize the partial likelihoods on the control and treatment group respectively, and obtain estimators $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$.
- (3) Estimation of the probability of not censored. We compute/estimate the not censored probabilities $\hat{\mathbb{P}}(C \geq Y \mid X = x, W = w)$ based on closed form solution, by numerical integral, or by classifiers with response Δ and predictors (X, W) ;
- (4) Substitution. We construct $\hat{a}(x)$ and $\hat{\nu}(x)$ based on $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$, $\hat{e}(x)$, and $\hat{\mathbb{P}}(C \geq Y \mid X = x, W = w)$.

2. On fold two, we plug in $\hat{a}(x)$ and $\hat{\nu}(x)$ to estimate $\tau(x)$ by maximizing the partial log-likelihood,

$$\min_{\tau'(x)} \text{pl}_n(\tau'(x); \hat{a}(x), \hat{\nu}(x)) := \frac{1}{n} \sum_{\Delta_i=1} (\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\} \tau'(X_i) - \log \left(\sum_{\mathcal{R}_i} e^{\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\} \tau'(X_i)} \right)), \quad (36)$$

where $\mathcal{R}_i = \{j : Y_j^c \geq Y_i^c\}$ denotes the risk set of subject i .

3. We swap the folds and obtain another estimate. We average the two estimates and output.

We start with 9 continuous lab measurements, follow the data preprocessing procedures of (Powers et al., 2018), and end up with 7517 units. Since SBP and DBP, EGFR and SCREAT are seriously correlated, respectively, we further remove DBP and SCREAT and are left with 7 covariates³.

5.1.3 Additional figures. We present the simulation results using gradient boosting to estimate the nuisance functions in Figure 1. The configurations of the hyperparameters α , β , θ , and δ are identical to those used in the simulations of the main text.

In order to demonstrate the robustness of the proposed method to inaccurate nuisance-

³In practice, we recommend selecting the target estimand based on both modeling considerations and scientific background, with priority given to the latter. Scientific knowledge is essential for supporting the unconfoundedness assumption in observational studies. When multiple modeling choices are equally plausible from a practitioner's perspective, we suggest favoring the one with more desirable statistical properties—for example, those enabling more powerful inference.

function estimators, in gradient boosting, we consider a small number (50) of trees, learning rate 0.1, and tree-depth 3. We limit the number of trees and the nuisance-function estimators will underfit the data. The results show that the proposed estimator is the most robust to inaccurate nuisance-function estimators.

[Figure 1 about here.]

5.1.4 *Additional tables.* We display the coverage and width of the 95% confidence intervals for various types of responses and nuisance-function estimators.

We use bootstrap (100 bootstrap samples) to construct confidence intervals for $\hat{\beta}$. We remark that Wald-type inference approaches based on the asymptotic distribution of the causal parameter is available (Chernozhukov et al., 2018). However, those methods require the nuisance function estimators to be of $o(n^{-1/4})$ accuracy and are only valid asymptotically. Given that we learn the nuisance functions by gradient boosting which does not guarantee to meet the desired criteria of the nuisance function estimation, we use bootstrap instead of the Wald tests for constructing confidence intervals.

- Table 2 considers Gaussian responses and uses linear regression as the nuisance-function estimator.
- Table 3 considers binary responses and uses logistic regression as the nuisance-function estimator.
- Table 4 considers Poisson responses and uses Poisson regression as the nuisance-function estimator.
- Table 5 considers survival responses and uses Cox regression as the nuisance-function estimator.
- Table 6 considers Gaussian responses and uses gradient boosting as the nuisance-function estimator.

- Table 7 considers binary responses and uses gradient boosting as the nuisance-function estimator.
- Table 8 considers Poisson responses and uses gradient boosting as the nuisance-function estimator.
- Table 9 considers survival responses and uses gradient boosting as the nuisance-function estimator.

We observe the proposed method (DINA) produces the best coverages for all $\hat{\beta}$ and sample sizes. The propensity score adjusted X-learner (PA-X) produces the second-best coverages but requires wider confidence intervals. For the direct extension of R-learner (E) (except for Gaussian responses), X-learner (X), and separate estimation (SE), the coverages of confidence intervals (for $\hat{\beta}_1$ in particular) decrease as the sample size grows. The phenomenon is due to the non-vanishing bias in those estimators.

[Table 2 about here.]

[Table 3 about here.]

[Table 4 about here.]

[Table 5 about here.]

[Table 6 about here.]

[Table 7 about here.]

[Table 8 about here.]

[Table 9 about here.]

5.2 Real data analysis

5.2.1 *Additional table.* In Table 10, we provide the descriptive statistics of the SPRINT data after preprocessing.

[Table 10 about here.]

5.2.2 *Analysis of sensitivity to treatment assignment.* We compare the sensitivity to treatment assignment mechanisms. We generate subsamples from the original dataset with non-constant propensity score and compare the estimated treatment effects from the original randomized trial and those from the hypothetical observational data.

Let $e(x)$ be an hypothetical propensity score following a logistic regression model. We subsample from the original SPRINT dataset with probabilities

$$\mathbb{P}(\text{select the } i\text{-th sample} \mid X_i = x) = \begin{cases} \min \left\{ \frac{e(x)}{1-e(x)}, 1 \right\}, & W_i = 1, \\ \min \left\{ \frac{1-e(x)}{e(x)}, 1 \right\}, & W_i = 0. \end{cases}$$

In the artificial subsamples, the probability of a unit being treated is $e(x)$. We run five estimation methods on the hypothetical subsamples, obtain estimated treatment effects, and repeat from subsampling 100 times. For nuisance-function estimation, we use logistic regression to learn the propensity score, and random forests or logistic regression to learn the treatment/control natural parameter functions.

For any method, we regard the estimates obtained from the original dataset as the oracle, denoted by τ_{method} , and compare with the estimates obtained from hypothetical subsamples, denoted by $\hat{\tau}_{\text{method}}$. We quantify the similarity by one minus the normalized mean squared difference,

$$R^2(\hat{\tau}_{\text{method}}) := 1 - \frac{\sum_{i=1}^n \{\hat{\tau}_{\text{method}}(X_i) - \tau_{\text{method}}(X_i)\}^2}{\sum_{i=1}^n \tau_{\text{method}}^2(X_i)}. \quad (37)$$

The larger $R^2(\hat{\tau}_{\text{method}})$ is, the closer $\hat{\tau}_{\text{method}}$ is to its oracle τ_{method} . In other words, a method with a larger $R^2(\hat{\tau}_{\text{method}})$ is more robust to the experimental design.

From Table 11 we observe the proposed method (DINA) produces the largest or the second-largest R^2 . The advantage is more evident when we use generalized linear regression as the nuisance-function learner.

[Table 11 about here.]

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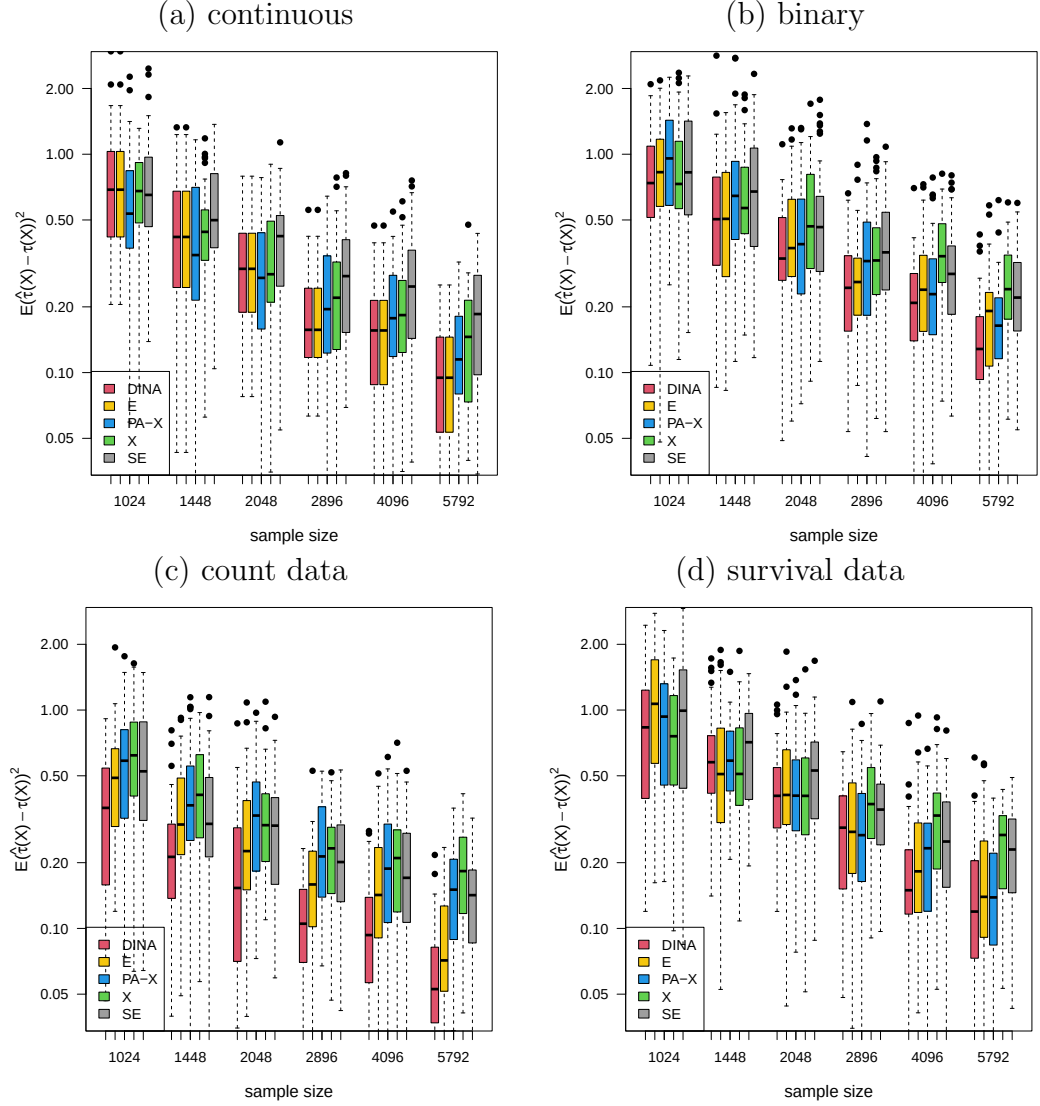


Figure 1. Estimation error log-log boxplots. We display the estimation errors $\mathbb{E}[(\hat{\tau}(X) - \tau(X))^2]$ over sample sizes in $[1024, 1448, 2048, 2896, 4096, 5792]$. We compare five methods: separate estimation (SE), X-learner (X), propensity score adjusted X-learner (PA-X), direct extension of R-learner (E), and the proposed method (DINA). We adopt the response model in the simulation section and consider four types of responses: (a) continuous (Gaussian); (b) binary (Bernoulli); (c) count data (Poisson); (d) survival data (Cox model with uniform censoring, 75% units censored). We estimate the propensity score by logistic regression, and estimate the natural parameters by gradient boosting. We repeat all experiments 100 times.

Table 1*Examples of censoring mechanisms and the associated probabilities of “not censored”.*

Censoring mechanism	Parameters	Density of censoring times $f_C(c)$	Probability of “not censored” $\mathbb{P}(C \geq Y \mid W = w, X = x)$
No censoring	-	0	1
Singly-censored	T	$\delta_{\{c=T\}}$	$1 - e^{-\Lambda(T)\eta_w(x)}$
Multiply-censored	Uniform	$1/T$	$1 - \frac{1}{T} \int_0^T e^{-\Lambda(c)\eta_w(x)} dc$
	Weibull	r, k	$1 - kr \int_0^\infty e^{-\Lambda(c)\eta_w(x)} (rc)^{k-1} e^{-(rc)^k} dc$

Table 2

Coverage (cvrg) and width of 95% confidence intervals for Gaussian responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. For Gaussian responses, the proposed method (DINA) and R-learner (E) are the same. We estimate the propensity score by logistic regression and baseline natural parameter functions by linear regression. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.97	1.48	0.96	1.49	0.93	1.38	0.95	1.35	0.93	1.34	0.98	0.789
	E	0.97	1.48	0.96	1.49	0.93	1.38	0.95	1.35	0.93	1.34	0.98	0.789
	PA-X	0.97	1.56	0.93	1.56	0.91	1.46	0.94	1.44	0.96	1.47	0.97	0.787
	X	0.50	1.35	0.48	1.36	0.91	1.29	0.96	1.27	0.95	1.25	0.99	0.811
	SE	0.46	1.36	0.53	1.34	0.87	1.26	0.91	1.27	0.91	1.28	0.92	0.795
1448	DINA	0.96	1.26	0.96	1.26	0.96	1.10	0.95	1.13	0.92	1.13	0.91	0.660
	E	0.96	1.26	0.96	1.26	0.96	1.10	0.95	1.13	0.92	1.13	0.91	0.660
	PA-X	0.97	1.30	0.97	1.29	0.93	1.22	0.97	1.20	0.98	1.22	0.95	0.653
	X	0.31	1.14	0.37	1.15	0.96	1.08	0.95	1.06	0.94	1.08	0.94	0.667
	SE	0.35	1.14	0.40	1.17	0.97	1.08	0.92	1.05	0.94	1.08	0.97	0.669
2048	DINA	0.93	1.04	0.94	1.04	0.94	0.957	0.95	0.955	0.94	0.947	0.93	0.549
	E	0.93	1.04	0.94	1.04	0.94	0.957	0.95	0.955	0.94	0.947	0.93	0.549
	PA-X	0.93	1.07	0.95	1.05	0.95	1.00	0.96	1.01	0.99	0.991	0.95	0.544
	X	0.25	0.975	0.24	0.946	0.95	0.891	0.97	0.886	0.95	0.898	0.94	0.576
	SE	0.24	0.950	0.32	0.949	0.97	0.883	0.94	0.897	0.94	0.871	0.94	0.562
2896	DINA	0.93	0.869	0.94	0.884	0.94	0.796	0.96	0.794	0.98	0.789	0.93	0.454
	E	0.93	0.869	0.94	0.884	0.94	0.796	0.96	0.794	0.98	0.789	0.93	0.454
	PA-X	0.96	0.906	0.98	0.901	0.93	0.847	0.94	0.840	0.95	0.846	0.97	0.462
	X	0.10	0.810	0.09	0.803	0.96	0.750	0.93	0.743	0.95	0.729	0.96	0.472
	SE	0.13	0.833	0.12	0.822	0.95	0.748	0.95	0.751	0.94	0.761	0.93	0.466
4096	DINA	0.99	0.733	0.94	0.754	0.91	0.669	0.96	0.659	0.92	0.662	0.93	0.389
	E	0.99	0.733	0.94	0.754	0.91	0.669	0.96	0.659	0.92	0.662	0.93	0.389
	PA-X	0.96	0.749	0.96	0.739	0.94	0.705	0.99	0.685	0.95	0.707	0.98	0.380
	X	0.01	0.685	0.04	0.682	0.92	0.629	0.97	0.642	0.97	0.627	0.97	0.397
	SE	0.05	0.675	0.06	0.669	0.96	0.630	0.96	0.625	0.93	0.632	0.90	0.395
5792	DINA	0.95	0.618	0.95	0.621	0.93	0.559	0.96	0.556	0.96	0.561	0.97	0.325
	E	0.95	0.618	0.95	0.621	0.93	0.559	0.96	0.556	0.96	0.561	0.97	0.325
	PA-X	0.97	0.627	0.98	0.627	0.97	0.595	0.90	0.575	0.96	0.597	0.94	0.321
	X	0.01	0.563	0.00	0.566	0.91	0.524	0.93	0.524	0.98	0.535	0.93	0.334
	SE	0.01	0.573	0.00	0.557	0.88	0.519	0.93	0.528	0.95	0.521	0.95	0.332

Table 3

Coverage (cvrg) and width of 95% confidence intervals for binary responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by logistic regression. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.93	1.67	0.94	1.65	0.93	1.59	0.97	1.52	0.98	1.51	0.97	0.875
	E	0.91	1.62	0.95	1.65	0.92	1.55	0.97	1.48	0.96	1.48	0.94	0.891
	PA-X	0.93	2.24	0.97	2.07	0.97	1.90	0.93	1.86	0.96	1.82	0.99	1.08
	X	0.45	1.49	0.85	1.57	0.96	1.46	0.93	1.42	0.99	1.41	0.95	0.930
	SE	0.47	1.47	0.76	1.53	0.97	1.42	0.96	1.40	0.97	1.39	0.96	0.931
1448	DINA	0.95	1.37	0.96	1.38	0.97	1.30	0.96	1.25	0.96	1.26	0.89	0.727
	E	0.88	1.31	0.92	1.36	0.98	1.25	0.97	1.23	0.97	1.24	0.89	0.715
	PA-X	1.00	1.75	0.96	1.59	0.96	1.54	0.96	1.46	0.98	1.49	0.98	0.887
	X	0.26	1.23	0.80	1.27	0.92	1.19	0.96	1.15	0.97	1.16	0.93	0.767
	SE	0.30	1.22	0.79	1.30	0.92	1.20	0.96	1.16	0.92	1.16	0.90	0.760
2048	DINA	0.95	1.14	0.95	1.14	0.95	1.11	0.95	1.04	0.95	1.04	0.86	0.619
	E	0.86	1.08	0.85	1.12	0.98	1.05	0.95	1.03	0.94	1.03	0.82	0.599
	PA-X	0.95	1.41	0.92	1.27	0.96	1.21	0.93	1.21	0.97	1.20	0.95	0.693
	X	0.14	1.02	0.78	1.06	0.95	1.00	0.96	0.950	0.97	0.943	0.80	0.616
	SE	0.18	1.03	0.71	1.06	0.95	0.993	0.97	0.953	0.98	0.961	0.85	0.621
2896	DINA	0.95	0.941	0.87	0.921	0.93	0.902	0.97	0.872	0.95	0.867	0.95	0.503
	E	0.82	0.896	0.79	0.931	0.90	0.866	0.96	0.832	0.95	0.857	0.84	0.494
	PA-X	0.92	1.10	0.93	1.03	0.93	0.990	0.94	0.973	0.94	0.980	0.94	0.565
	X	0.06	0.845	0.54	0.881	0.92	0.814	0.94	0.795	0.95	0.797	0.84	0.524
	SE	0.05	0.847	0.62	0.901	0.92	0.836	0.95	0.801	0.97	0.803	0.89	0.523
4096	DINA	0.92	0.77	0.93	0.777	0.96	0.728	0.95	0.708	0.97	0.715	0.89	0.424
	E	0.77	0.762	0.8	0.764	0.96	0.727	0.96	0.705	0.95	0.703	0.66	0.410
	PA-X	0.95	0.921	0.84	0.849	0.92	0.827	0.94	0.807	0.94	0.813	0.95	0.457
	X	0.00	0.705	0.54	0.734	0.92	0.692	0.97	0.674	0.97	0.668	0.80	0.437
	SE	0.01	0.694	0.52	0.732	0.92	0.691	0.94	0.664	0.97	0.671	0.79	0.428
5792	DINA	0.92	0.647	0.85	0.658	0.96	0.638	0.97	0.608	0.97	0.600	0.82	0.358
	E	0.73	0.633	0.66	0.648	0.96	0.611	0.97	0.582	0.94	0.589	0.69	0.351
	PA-X	0.91	0.768	0.87	0.713	0.98	0.701	0.99	0.674	0.91	0.659	0.89	0.381
	X	0.00	0.606	0.41	0.624	0.97	0.565	0.93	0.564	0.97	0.568	0.65	0.368
	SE	0.00	0.594	0.39	0.624	0.92	0.575	0.97	0.558	0.95	0.552	0.68	0.360

Table 4

Coverage (cvrg) and width of 95% confidence intervals for count data. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by Poisson regression. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.92	0.679	0.92	0.716	0.93	0.646	0.93	0.631	0.96	0.642	0.94	0.379
	E	0.57	0.706	0.48	0.732	0.97	0.661	0.93	0.639	0.94	0.640	0.93	0.452
	PA-X	0.94	0.880	0.89	0.91	0.97	0.669	0.95	0.666	0.94	0.666	0.93	0.419
	X	0.08	0.636	0.94	0.691	0.95	0.597	0.92	0.603	0.91	0.584	0.91	0.417
	SE	0.03	0.628	0.94	0.698	0.99	0.599	0.93	0.591	0.93	0.591	0.94	0.421
1448	DINA	0.96	0.553	0.92	0.580	0.91	0.546	0.95	0.534	0.91	0.526	0.94	0.314
	E	0.44	0.569	0.45	0.605	0.91	0.540	0.98	0.536	0.94	0.524	0.96	0.372
	PA-X	0.87	0.734	0.86	0.746	0.95	0.554	0.93	0.552	0.93	0.544	0.95	0.339
	X	0.00	0.536	0.93	0.573	0.93	0.495	0.93	0.500	0.96	0.496	0.87	0.349
	SE	0.04	0.532	0.95	0.587	0.97	0.500	0.94	0.492	0.94	0.499	0.93	0.349
2048	DINA	0.97	0.478	0.91	0.491	0.91	0.450	0.90	0.450	0.96	0.449	0.95	0.271
	E	0.37	0.485	0.30	0.519	0.91	0.462	0.93	0.445	0.96	0.446	0.96	0.314
	PA-X	0.87	0.589	0.73	0.636	0.96	0.466	0.95	0.458	0.91	0.449	0.93	0.282
	X	0.00	0.447	0.93	0.486	0.95	0.422	0.95	0.424	0.93	0.422	0.91	0.297
	SE	0.01	0.435	0.92	0.483	0.98	0.425	0.99	0.418	0.95	0.413	0.96	0.292
2896	DINA	0.89	0.393	0.91	0.413	0.98	0.378	0.92	0.370	0.93	0.370	0.97	0.225
	E	0.14	0.402	0.09	0.429	0.93	0.382	0.95	0.374	0.98	0.380	0.92	0.263
	PA-X	0.72	0.493	0.73	0.513	0.95	0.386	0.97	0.379	0.96	0.378	0.95	0.237
	X	0.00	0.378	0.97	0.410	0.93	0.347	0.95	0.348	0.96	0.349	0.91	0.245
	SE	0.00	0.383	0.95	0.404	0.94	0.358	0.93	0.349	0.93	0.347	0.97	0.243
4096	DINA	0.97	0.339	0.86	0.343	0.88	0.317	0.96	0.314	0.95	0.315	0.96	0.186
	E	0.02	0.332	0.06	0.364	0.91	0.317	0.97	0.318	0.94	0.311	0.90	0.213
	PA-X	0.65	0.411	0.64	0.430	0.98	0.323	0.93	0.312	0.96	0.315	0.91	0.200
	X	0.00	0.312	0.91	0.339	0.99	0.292	0.95	0.293	0.97	0.290	0.86	0.205
	SE	0.00	0.315	0.90	0.344	0.98	0.293	0.96	0.292	0.96	0.296	0.86	0.206
5792	DINA	0.95	0.276	0.94	0.291	0.92	0.269	0.94	0.260	0.95	0.261	0.93	0.158
	E	0.00	0.283	0.01	0.306	0.88	0.266	0.94	0.267	0.94	0.262	0.86	0.182
	PA-X	0.52	0.348	0.51	0.367	0.93	0.267	0.93	0.264	0.95	0.260	0.87	0.166
	X	0.00	0.266	0.91	0.287	0.91	0.245	0.94	0.247	0.95	0.247	0.84	0.172
	SE	0.00	0.263	0.86	0.288	0.97	0.254	0.96	0.245	0.92	0.246	0.84	0.173

Table 5

Coverage (cvrg) and width of 95% confidence intervals for survival responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by Cox regression. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.92	1.84	0.93	1.67	0.96	1.61	0.97	1.55	0.97	1.56	0.95	1.09
	E	0.87	1.91	0.91	1.66	0.96	1.58	0.96	1.54	0.92	1.51	0.92	1.11
	PA-X	0.95	3.38	0.98	3.23	0.97	2.14	1.00	2.01	0.97	2.04	0.97	1.20
	X	0.97	1.62	0.75	1.53	0.93	1.40	0.96	1.41	0.97	1.38	0.95	0.971
	SE	0.95	1.55	0.67	1.49	0.91	1.40	0.96	1.39	0.97	1.39	0.86	0.868
1448	DINA	0.92	1.52	0.88	1.33	0.93	1.29	0.97	1.29	0.94	1.27	0.90	0.885
	E	0.77	1.55	0.94	1.36	0.96	1.27	0.96	1.25	0.93	1.27	0.88	0.879
	PA-X	0.94	2.35	0.97	2.25	0.99	1.55	0.95	1.51	0.98	1.53	0.93	0.865
	X	0.88	1.30	0.43	1.25	0.94	1.15	0.94	1.15	0.91	1.11	0.94	0.770
	SE	0.89	1.29	0.51	1.26	0.95	1.15	0.93	1.15	0.95	1.15	0.83	0.718
2048	DINA	0.96	1.24	0.95	1.11	0.93	1.06	0.93	1.05	0.91	1.04	0.94	0.720
	E	0.73	1.28	0.94	1.11	0.89	1.04	0.95	1.03	0.91	1.02	0.96	0.734
	PA-X	0.92	1.77	0.94	1.68	0.97	1.20	0.95	1.19	0.98	1.18	0.94	0.672
	X	0.92	1.08	0.25	1.02	0.91	0.954	0.95	0.943	0.94	0.963	0.94	0.639
	SE	0.86	1.07	0.27	1.01	0.97	0.947	0.93	0.944	0.95	0.941	0.74	0.589
2896	DINA	0.93	1.04	0.88	0.926	0.93	0.868	0.95	0.866	0.94	0.873	0.90	0.586
	E	0.72	1.05	0.91	0.939	0.94	0.869	0.97	0.855	0.93	0.855	0.96	0.595
	PA-X	0.91	1.43	0.96	1.34	0.93	0.964	0.97	0.946	0.93	0.954	0.96	0.543
	X	0.93	0.903	0.19	0.838	0.94	0.799	0.95	0.776	0.94	0.778	0.96	0.530
	SE	0.88	0.881	0.19	0.844	0.95	0.785	0.90	0.772	0.95	0.788	0.56	0.468
4096	DINA	0.98	0.869	0.93	0.770	0.99	0.737	0.97	0.714	0.96	0.703	0.93	0.501
	E	0.48	0.872	0.91	0.777	0.99	0.724	0.96	0.715	0.97	0.717	0.93	0.486
	PA-X	0.92	1.14	0.94	1.06	0.98	0.805	0.97	0.786	0.98	0.772	0.92	0.453
	X	0.90	0.758	0.05	0.711	1.00	0.667	0.97	0.677	0.98	0.663	0.93	0.439
	SE	0.84	0.742	0.04	0.706	0.90	0.661	0.97	0.654	0.96	0.652	0.51	0.400
5792	DINA	0.93	0.715	0.93	0.633	0.93	0.605	0.94	0.594	0.92	0.595	0.93	0.410
	E	0.39	0.720	0.96	0.646	0.93	0.600	0.95	0.585	0.91	0.584	0.91	0.408
	PA-X	0.91	0.904	0.93	0.868	0.95	0.638	0.97	0.629	0.96	0.650	0.97	0.372
	X	0.85	0.623	0.02	0.590	0.94	0.546	0.95	0.541	0.98	0.546	0.99	0.369
	SE	0.82	0.619	0.00	0.590	0.97	0.549	0.94	0.542	0.94	0.541	0.30	0.327

Table 6

Coverage (cvrg) and width of 95% confidence intervals for Gaussian responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. For Gaussian responses, the proposed method (DINA) and R-learner (E) are the same. We estimate the propensity score by logistic regression and baseline natural parameter functions by gradient boosting. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.96	1.39	0.97	1.40	0.94	1.29	0.97	1.30	0.89	1.30	0.92	0.746
	E	0.96	1.39	0.97	1.40	0.94	1.29	0.97	1.30	0.89	1.30	0.92	0.746
	PA-X	0.93	1.43	0.86	1.35	0.92	1.26	0.99	1.21	0.95	1.20	0.95	0.787
	X	0.87	1.34	0.75	1.40	0.95	1.27	0.96	1.18	0.94	1.22	0.94	0.784
	SE	0.80	1.28	0.92	1.27	0.92	1.20	0.93	1.07	1.00	1.06	0.95	0.778
1448	DINA	0.99	1.16	0.91	1.14	0.93	1.09	0.94	1.11	0.88	1.11	0.94	0.617
	E	0.99	1.16	0.91	1.14	0.93	1.09	0.94	1.11	0.88	1.11	0.94	0.617
	PA-X	0.97	1.18	0.92	1.14	0.87	1.06	0.93	0.998	0.91	0.996	0.97	0.647
	X	0.89	1.12	0.81	1.19	0.93	1.03	0.96	1.01	0.91	1.02	0.93	0.650
	SE	0.85	1.04	0.89	1.05	0.91	0.968	0.96	0.88	0.91	0.887	0.92	0.650
2048	DINA	0.93	0.956	0.95	0.954	0.94	0.933	0.91	0.917	0.93	0.919	0.97	0.530
	E	0.93	0.956	0.95	0.954	0.94	0.933	0.91	0.917	0.93	0.919	0.97	0.530
	PA-X	0.96	0.992	0.89	0.965	0.91	0.900	0.90	0.837	0.94	0.821	0.95	0.546
	X	0.82	0.927	0.74	0.970	0.90	0.888	0.93	0.84	0.97	0.821	0.96	0.547
	SE	0.77	0.889	0.89	0.889	0.88	0.858	0.94	0.716	0.94	0.727	0.89	0.531
2896	DINA	0.92	0.797	0.87	0.819	0.94	0.765	0.89	0.758	0.95	0.765	0.94	0.446
	E	0.92	0.797	0.87	0.819	0.94	0.765	0.89	0.758	0.95	0.765	0.94	0.446
	PA-X	0.91	0.831	0.85	0.808	0.88	0.755	0.90	0.687	0.93	0.688	0.93	0.450
	X	0.79	0.764	0.65	0.833	0.89	0.736	0.91	0.692	0.92	0.695	0.94	0.454
	SE	0.83	0.745	0.90	0.740	0.86	0.697	0.96	0.590	0.92	0.602	0.94	0.437
4096	DINA	0.95	0.682	0.86	0.686	0.96	0.638	0.95	0.648	0.98	0.639	0.94	0.367
	E	0.95	0.682	0.86	0.686	0.96	0.638	0.95	0.648	0.98	0.639	0.94	0.367
	PA-X	0.93	0.706	0.88	0.688	0.84	0.637	0.94	0.584	0.96	0.573	0.93	0.375
	X	0.83	0.649	0.72	0.711	0.88	0.624	0.95	0.581	0.96	0.576	0.92	0.382
	SE	0.67	0.629	0.90	0.628	0.91	0.597	0.96	0.496	0.95	0.495	0.95	0.373
5792	DINA	0.95	0.562	0.93	0.572	0.94	0.550	0.96	0.550	0.89	0.541	0.91	0.307
	E	0.95	0.562	0.93	0.572	0.94	0.550	0.96	0.550	0.89	0.541	0.91	0.307
	PA-X	0.89	0.589	0.88	0.576	0.87	0.538	0.96	0.476	0.92	0.476	0.95	0.311
	X	0.79	0.55	0.62	0.582	0.91	0.516	0.97	0.477	0.91	0.479	0.92	0.316
	SE	0.61	0.521	0.92	0.530	0.91	0.516	0.97	0.401	0.96	0.410	0.89	0.312

Table 7

Coverage (cvrg) and width of 95% confidence intervals for binary responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by gradient boosting. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.93	1.59	0.92	1.63	0.96	1.53	0.91	1.53	0.94	1.52	0.97	0.897
	E	0.91	1.65	0.96	1.70	0.96	1.61	0.92	1.58	0.91	1.55	0.95	0.913
	PA-X	0.91	1.80	0.91	1.86	0.93	1.70	0.95	1.58	0.97	1.59	0.92	1.10
	X	0.92	1.63	0.88	1.79	0.96	1.63	0.96	1.53	0.97	1.52	0.94	1.04
	SE	0.82	1.50	0.93	1.52	0.94	1.45	0.96	1.28	0.97	1.27	0.94	0.934
1448	DINA	0.89	1.30	0.95	1.37	0.90	1.24	0.93	1.22	0.93	1.22	0.92	0.728
	E	0.90	1.33	0.91	1.38	0.89	1.31	0.92	1.26	0.96	1.28	0.91	0.752
	PA-X	0.95	1.50	0.97	1.50	0.89	1.36	0.94	1.27	0.98	1.27	0.94	0.875
	X	0.90	1.33	0.96	1.48	0.88	1.33	0.95	1.22	0.96	1.23	0.95	0.825
	SE	0.74	1.21	0.97	1.27	0.91	1.20	0.96	1.04	0.96	1.04	0.89	0.776
2048	DINA	0.91	1.09	0.96	1.11	0.98	1.03	0.92	1.03	0.92	1.03	0.97	0.607
	E	0.89	1.10	0.88	1.11	0.98	1.08	0.93	1.04	0.91	1.04	0.97	0.608
	PA-X	0.97	1.21	0.88	1.22	0.87	1.14	0.94	1.05	0.92	1.04	0.99	0.708
	X	0.80	1.07	0.84	1.20	0.91	1.07	0.93	1.00	0.97	1.00	0.99	0.676
	SE	0.55	0.986	0.98	1.04	0.92	0.990	0.94	0.848	0.92	0.847	0.86	0.622
2896	DINA	0.97	0.913	0.94	0.925	0.95	0.866	0.89	0.859	0.93	0.859	0.88	0.506
	E	0.94	0.893	0.88	0.926	0.95	0.890	0.89	0.868	0.93	0.856	0.84	0.514
	PA-X	0.96	1.01	0.88	1.00	0.86	0.933	0.96	0.846	0.95	0.844	0.96	0.583
	X	0.77	0.907	0.86	1.00	0.90	0.877	0.97	0.840	0.97	0.817	0.96	0.567
	SE	0.50	0.842	0.95	0.867	0.93	0.811	0.97	0.678	0.90	0.682	0.86	0.522
4096	DINA	0.91	0.753	0.95	0.773	0.92	0.732	0.95	0.722	0.91	0.702	0.93	0.420
	E	0.85	0.757	0.86	0.761	0.93	0.734	0.92	0.711	0.96	0.730	0.88	0.422
	PA-X	0.96	0.841	0.91	0.826	0.84	0.765	0.92	0.676	0.96	0.689	0.96	0.464
	X	0.75	0.750	0.85	0.828	0.87	0.740	0.93	0.666	0.96	0.672	0.95	0.459
	SE	0.43	0.699	0.93	0.720	0.90	0.668	0.95	0.557	0.97	0.557	0.82	0.423
5792	DINA	0.94	0.635	0.97	0.646	0.97	0.622	0.96	0.610	0.94	0.607	0.91	0.363
	E	0.89	0.633	0.83	0.636	0.91	0.618	0.97	0.607	0.93	0.597	0.85	0.357
	PA-X	0.95	0.699	0.86	0.715	0.92	0.664	0.92	0.579	0.97	0.569	0.95	0.396
	X	0.61	0.619	0.81	0.686	0.96	0.615	0.91	0.544	0.94	0.550	0.94	0.379
	SE	0.33	0.582	0.95	0.619	0.84	0.569	0.96	0.450	0.91	0.444	0.78	0.359

Table 8

Coverage (cvrg) and width of 95% confidence intervals for count data. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by gradient boosting. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.95	0.712	0.91	0.769	0.91	0.694	0.95	0.670	0.91	0.676	0.92	0.392
	E	0.95	0.719	0.89	0.766	0.89	0.701	0.95	0.692	0.95	0.694	0.94	0.410
	PA-X	0.81	0.701	0.93	0.733	0.86	0.650	0.96	0.568	0.96	0.559	0.97	0.396
	X	0.84	0.666	0.92	0.722	0.90	0.648	0.97	0.569	0.95	0.568	0.95	0.384
	SE	0.82	0.594	0.82	0.637	0.77	0.576	0.97	0.487	0.94	0.493	0.92	0.387
1448	DINA	0.97	0.580	0.94	0.632	0.90	0.565	0.95	0.552	0.91	0.562	0.92	0.331
	E	0.98	0.608	0.96	0.628	0.95	0.583	0.95	0.569	0.92	0.551	0.93	0.339
	PA-X	0.89	0.581	0.86	0.601	0.75	0.547	0.93	0.462	0.98	0.462	0.92	0.332
	X	0.91	0.555	0.86	0.600	0.77	0.533	0.95	0.466	0.97	0.470	0.91	0.323
	SE	0.76	0.492	0.77	0.541	0.71	0.476	0.96	0.404	0.97	0.399	0.92	0.317
2048	DINA	0.92	0.484	0.95	0.523	0.94	0.485	0.93	0.476	0.95	0.480	0.91	0.271
	E	0.94	0.492	0.92	0.525	0.94	0.471	0.95	0.468	0.95	0.468	0.92	0.279
	PA-X	0.86	0.487	0.83	0.499	0.78	0.458	0.94	0.378	0.96	0.378	0.92	0.268
	X	0.89	0.462	0.84	0.502	0.87	0.462	0.93	0.384	0.98	0.387	0.91	0.275
	SE	0.69	0.413	0.77	0.445	0.73	0.396	0.95	0.325	0.92	0.329	0.96	0.265
2896	DINA	0.95	0.405	0.93	0.445	0.89	0.402	0.91	0.400	0.94	0.395	0.94	0.233
	E	0.96	0.407	0.92	0.437	0.92	0.400	0.92	0.390	0.96	0.387	0.92	0.239
	PA-X	0.79	0.403	0.80	0.425	0.81	0.395	0.94	0.320	0.96	0.314	0.95	0.224
	X	0.78	0.391	0.82	0.415	0.84	0.389	0.96	0.315	0.95	0.320	0.94	0.223
	SE	0.59	0.352	0.79	0.372	0.79	0.329	0.96	0.266	0.94	0.256	0.91	0.216
4096	DINA	0.97	0.344	0.93	0.37	0.94	0.342	0.94	0.338	0.89	0.329	0.93	0.195
	E	0.98	0.344	0.89	0.362	0.94	0.334	0.96	0.320	0.91	0.324	0.95	0.198
	PA-X	0.86	0.341	0.81	0.346	0.77	0.329	0.95	0.260	0.94	0.261	0.93	0.186
	X	0.89	0.331	0.86	0.357	0.81	0.324	0.93	0.263	0.96	0.261	0.91	0.187
	SE	0.51	0.301	0.70	0.304	0.63	0.281	0.94	0.213	0.94	0.213	0.91	0.182
5792	DINA	0.94	0.295	0.93	0.313	0.98	0.290	0.90	0.218	0.95	0.237	0.94	0.166
	E	0.96	0.287	0.91	0.308	0.95	0.280	0.92	0.212	0.94	0.231	0.89	0.162
	PA-X	0.78	0.291	0.79	0.299	0.76	0.278	0.96	0.216	0.92	0.214	0.95	0.157
	X	0.81	0.278	0.83	0.297	0.79	0.278	0.95	0.216	0.94	0.221	0.92	0.157
	SE	0.48	0.253	0.46	0.258	0.53	0.231	0.93	0.169	0.97	0.173	0.93	0.155

Table 9

Coverage (cvrg) and width of 95% confidence intervals for survival responses. We consider the five meta-algorithms and vary the sample size from 1024 to 5792. We estimate the propensity score by logistic regression and baseline natural parameter functions by gradient boosting. Confidence intervals are constructed using 100 bootstrap samples. Results are averaged over 100 trials.

Sample size	Method	β_1		β_2		β_3		β_4		β_5		β_0	
		cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width	cvrg	width
1024	DINA	0.95	1.86	0.95	1.74	0.94	1.69	0.96	1.63	0.99	1.63	0.95	1.05
	E	0.90	1.86	0.95	1.74	0.97	1.66	0.93	1.62	0.94	1.63	0.96	1.06
	PA-X	0.96	1.89	0.95	1.91	0.99	1.67	0.97	1.58	0.98	1.58	0.97	1.16
	X	0.91	1.72	0.87	1.62	0.97	1.61	0.98	1.57	0.97	1.58	0.94	1.12
	SE	0.88	1.49	0.90	1.43	0.97	1.43	0.97	1.38	0.98	1.35	0.88	0.894
1448	DINA	0.98	1.47	0.91	1.40	0.94	1.35	0.91	1.31	0.94	1.33	0.95	0.837
	E	0.95	1.46	0.91	1.39	0.98	1.30	0.91	1.31	0.98	1.31	0.92	0.837
	PA-X	0.91	1.49	0.93	1.56	0.93	1.33	0.96	1.24	0.94	1.26	0.98	0.901
	X	0.83	1.38	0.85	1.32	0.95	1.31	0.95	1.26	0.95	1.25	0.97	0.859
	SE	0.79	1.19	0.90	1.16	0.88	1.18	0.95	1.12	0.92	1.11	0.80	0.717
2048	DINA	0.91	1.23	0.95	1.16	0.97	1.07	0.98	1.06	0.95	1.05	0.91	0.663
	E	0.85	1.19	0.93	1.12	0.94	1.05	0.95	1.04	0.95	1.04	0.94	0.658
	PA-X	0.93	1.21	0.93	1.25	0.93	1.05	0.97	1.02	0.94	1.01	0.97	0.699
	X	0.78	1.14	0.77	1.05	0.96	1.06	0.99	1.01	0.93	0.997	0.97	0.675
	SE	0.71	1.02	0.86	0.943	0.92	0.927	0.90	0.905	0.94	0.901	0.70	0.582
2896	DINA	0.96	0.975	0.96	0.943	0.96	0.907	0.97	0.883	0.95	0.887	0.85	0.541
	E	0.90	0.973	0.94	0.917	0.95	0.879	0.94	0.838	0.94	0.856	0.93	0.534
	PA-X	0.86	0.995	0.94	1.02	0.88	0.855	0.99	0.817	0.91	0.815	0.93	0.549
	X	0.84	0.913	0.78	0.861	0.93	0.862	0.97	0.807	0.93	0.803	0.91	0.541
	SE	0.65	0.812	0.87	0.777	0.90	0.760	0.95	0.712	0.92	0.719	0.52	0.478
4096	DINA	0.95	0.831	0.88	0.768	0.97	0.747	0.95	0.732	0.88	0.719	0.91	0.452
	E	0.79	0.815	0.90	0.767	0.96	0.728	0.95	0.697	0.93	0.703	0.95	0.45
	PA-X	0.85	0.830	0.97	0.835	0.89	0.717	0.96	0.652	0.97	0.645	0.90	0.444
	X	0.71	0.770	0.78	0.737	0.92	0.707	0.95	0.669	0.93	0.648	0.92	0.437
	SE	0.57	0.684	0.81	0.651	0.85	0.642	0.95	0.580	0.98	0.595	0.48	0.406
5792	DINA	0.92	0.696	0.93	0.632	0.92	0.599	0.97	0.583	0.95	0.598	0.91	0.382
	E	0.81	0.675	0.89	0.620	0.94	0.594	0.98	0.577	0.97	0.575	0.96	0.375
	PA-X	0.76	0.669	0.94	0.675	0.85	0.602	0.97	0.521	0.97	0.531	0.90	0.354
	X	0.60	0.625	0.74	0.585	0.92	0.573	0.94	0.530	0.98	0.536	0.93	0.359
	SE	0.48	0.569	0.91	0.529	0.80	0.518	0.95	0.478	0.97	0.483	0.24	0.331

Table 10
Descriptive statistics of the SPRINT data after preprocessing. Standard deviations are in parentheses.

Type	Abbreviation	Feature	Treatment Group	Control Group
Group size	—	—	3708	3809
Lab meas- urements	SBP	Seated systolic blood pressure (mm Hg)	139 (15.6)	140 (15.5)
	DBP	Seated diastolic blood pressure (mm Hg)	78.3 (11.7)	78.2 (12.1)
	EGFR	eGFR MDRD (mL/min/1.73m ²)	72.0 (20.5)	72.1 (20.2)
	SCREAT	Serum creatinine (mg/dL)	1.07 (0.34)	1.07 (0.32)
	CHR	Cholesterol (mg/dL)	191 (41.7)	190 (40.4)
	GLUR	Glucose (mg/dL)	99.0 (13.7)	98.9 (13.3)
	HDL	HDL-cholesterol direct (mg/dL)	52.7 (14.3)	52.7 (14.5)
	TRR	Triglycerides (mg/dL)	126 (22.2)	126 (19.6)
	UMALCR	Urine Albumin/Creatinine ratio (mg/g)	39.8 (15.8)	36.6 (12.5)

Table 11

Comparison of the sensitivity (R^2 in (37)) to treatment assignment mechanisms. We consider the five methods in the simulation section in the main text. We generate 100 subsamples from the original SPRINT dataset with a non-constant propensity score. We consider two nuisance-function learners: logistic regression and random forests. For each method, we obtain estimators from the original SPRINT dataset and the hypothetical subsamples, and compute R^2 .

Nuisance-function learner	DINA	E	PA-X	X	SE
Logistic regression	0.501	0.387	0.107	0.0982	-0.468
	(0.186)	(0.252)	(0.522)	(0.404)	(0.772)
Random forests	0.436	0.242	0.549	0.426	-1.44
	(0.400)	(0.458)	(0.305)	(0.328)	(2.48)